

Model Comparison: gemini-3 vs gemma-4

Date: 2026-05-02

This document compares gemini-3 and gemma-4 using 2 independent evaluation sources. Each source may evaluate different model variants, so the sections should be read as complementary views rather than direct cross-references.

Section	Models Found	Evaluation Method
Part 1: Arena	gemini-3-flash, gemini-3-flash (thinking-minimal), gemini-3-pro, gemini-3.1-flash-lite-preview, gemini-3.1-pro-preview, gemma-4-26b-a4b, gemma-4-31b	Human Preference Ratings
Part 2: Artificial Analysis	Gemini 3 Flash Preview (Non-reasoning), Gemini 3.1 Pro Preview, Gemini 3 Pro Preview (low), Gemini 3.1 Flash-Lite Preview, Gemini 3 Flash Preview (Reasoning), Gemini 3 Pro Preview (high), Gemma 4 26B A4B (Reasoning), Gemma 4 31B (Reasoning), Gemma 4 26B A4B (Non-reasoning), Gemma 4 31B (Non-reasoning), Gemma 4 E2B (Non-reasoning), Gemma 4 E4B (Reasoning), Gemma 4 E4B (Non-reasoning), Gemma 4 E2B (Reasoning)	Automated Benchmarks

Overall Conclusions

- Overall positioning:** Both families are Google-built, but they serve fundamentally different roles. Gemini 3 is a proprietary API product line — its top model, gemini-3.1-pro-preview, sits at **rank 4 of 354** on Arena (Frontier tier, rating 1492.9) and **rank 3 of 503** on AA (Frontier tier, Intelligence Index 57). Gemma 4 is an open-weights research family — its top model, gemma-4-31b, ranks **38th on Arena** (Upper-mid tier, 1450.9) and scores 39 on AA’s Intelligence Index in reasoning mode (Upper-mid tier). The Arena gap between these tops is 42 points, which represents a clear separation by confidence interval analysis, though both are comfortably in the top 15% of all models globally.
- Lineup depth:** Gemini 3 fields 5 models on Arena and 6 configurations on AA (spanning Pro, Flash, and Flash-Lite variants with reasoning/non-reasoning modes). Gemma 4 has 2 models on Arena (31B and 26B-A4B) but 8 configurations on AA thanks to its broader size range (E2B, E4B, 26B, 31B — each in reasoning and non-reasoning modes). Gemma’s breadth is in model sizes; Gemini’s is in capability tiers.
- Value proposition:** Gemini 3 dominates on raw quality — sweeping all 8 Arena categories in every head-to-head matchup. Its Flash variant (rank 16, rating 1473.6) offers Near-frontier performance at \$1.12/1M blended tokens and 179.9 t/s. Gemma 4 competes on **cost and openness** — the 31B model delivers Upper-mid quality at \$0.20/1M blended tokens (5.6x cheaper than Gemini Flash), and as open-weights models, Gemma variants can be self-hosted and fine-tuned.
- Quality profile differences:** In Arena head-to-heads, Gemini 3.1 Pro Preview’s largest advantage over the Gemma models is in **creative writing** (+84.9 vs gemma-4-26b-a4b, +67.0 vs gemma-4-31b). Gemma’s strongest relative showing is in **math** and **coding** — the gap narrows to +25.8 and +31.1 respectively against the 31B model, and gemma-4-26b-a4b actually scores closer on ind_math (+20.4). The Gemma models appear to be more STEM-focused while Gemini has a broader, more humanities-leaning advantage.
- Evaluation coverage:** All models here are from Google, so the “head-to-head” comparisons are effectively intra-family. Arena coverage is limited for Gemma (only 2 models, ~5,800 votes each vs 27,000–41,000 for Gemini models). AA has more Gemma variants but is missing data for many fields (speed, latency, math index) on the smaller Gemma models. Context window data is unavailable for all models in this report.

6. Summary table:

Factor	Gemini 3	Gemma 4
Top-tier quality	Frontier (rank 4/354 Arena, rank 3/503 AA)	Upper-mid (rank 38/354 Arena)
Same-tier quality	gemini-3-pro (rank 8) ~ Near-frontier	gemma-4-31b (rank 38) ~ Upper-mid
Speed	126–185 t/s (Pro–Flash)	35.3 t/s (31B Reasoning)
Latency (TTFT)	0.70s (Flash non-reasoning) – 21.8s (Pro)	1.00s (31B Reasoning)
Price	\$1.12–\$4.50/1M blended	\$0.20/1M blended
Context window	Not reported	Not reported
Strength categories	Creative writing, instruction following	Math, coding (smaller relative gap)
Weakness categories	None — wins all categories	Creative writing, multi-turn
Model variety	5 Arena / 6 AA configs	2 Arena / 8 AA configs
Open weights	No (proprietary API)	Yes

7. **Bottom line:** Choose Gemini 3 when you need the highest quality available — its Pro variant competes with the very best models globally, and Flash offers a strong quality/cost tradeoff for high-volume applications. Choose Gemma 4 when you need open weights for fine-tuning, self-hosting, or edge deployment, or when cost is the primary constraint — the 31B model delivers credible performance at a fraction of Gemini’s price, particularly for STEM-oriented tasks.

Part 1: Arena

Source: Arena

About Arena

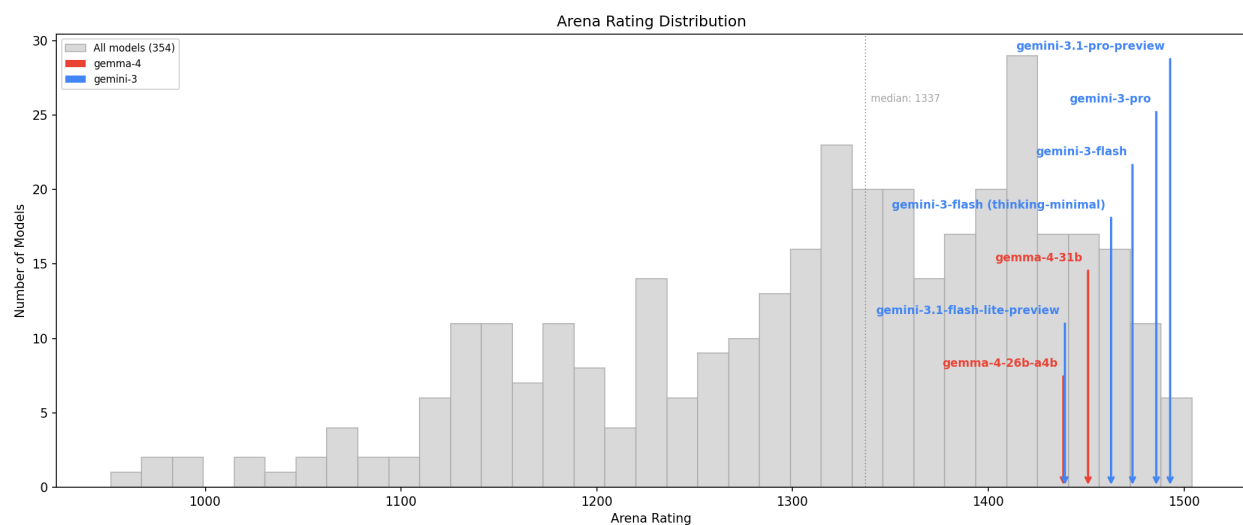
Arena (formerly LMSYS Chatbot Arena) ranks models using **head-to-head human preference votes**. Real users submit prompts to two anonymous models side-by-side, then choose which response they prefer. These pairwise outcomes are aggregated into Bradley-Terry ratings (similar to Elo in chess) – higher is better, with scores typically ranging from ~900 to ~1550.

The `text_style_control` variant adjusts for verbosity bias, so models don’t get rewarded simply for producing longer responses.

Ratings are broken down into **27 topic-based categories** (coding, math, creative writing, legal, etc.) based on conversation content, giving a detailed profile of where each model excels relative to the full field. Because these are human judgments rather than automated test suites, they reflect what real users find helpful – but they also carry the biases of the voter population (skewed toward AI-savvy early adopters) and the preference-vs-correctness gap (a confident but wrong answer can still win votes).

Models evaluated: gemini-3-flash, gemini-3-flash (thinking-minimal), gemini-3-pro, gemini-3.1-flash-lite-preview, gemini-3.1-pro-preview, gemma-4-26b-a4b, gemma-4-31b

Global Arena Rankings (354 models total)



Rank	Model	Rating	Votes
1	claude-opus-4-7-thinking	1504.0	7,467
2	claude-opus-4-6-thinking	1502.4	22,240
3	claude-opus-4-6	1496.8	23,704
4	gemini-3.1-pro-preview	1492.9	27,941
5	claude-opus-4-7	1492.6	8,206
6	muse-spark	1490.5	9,278
7	gpt-5.5-high	1488.1	4,963
8	gemini-3-pro	1485.8	41,373
9	grok-4.20-beta1	1479.8	17,253
10	gpt-5.4-high	1477.3	15,705
11	grok-4.20-beta-0309-reasoning	1477.2	16,048
12	gpt-5.2-chat-latest-20260210	1476.8	22,155
13	grok-4.20-multi-agent-beta-0309	1475.0	16,307
14	ernie-5.1-preview	1474.3	4,609
15	gpt-5.5	1474.0	5,153
16	gemini-3-flash	1473.6	30,800
17	claude-opus-4-5-20251101-thinking-32k	1472.7	37,170
18	glm-5.1	1470.3	10,949
19	grok-4.1-thinking	1468.5	53,794
20	claude-opus-4-5-20251101	1468.0	53,446
21	gpt-5.4	1465.5	16,407
22	claude-sonnet-4-6	1463.7	15,657
23	qwen3.5-max-preview	1463.6	13,419
24	mimo-v2.5-pro	1463.6	4,901
25	deepseek-v4-pro	1463.3	4,175
26	gemini-3-flash (thinking-minimal)	1462.7	39,847
27	kimi-k2.6	1462.1	6,067
28	deepseek-v4-pro-thinking	1461.9	3,816
29	grok-4.1	1460.2	57,733
30	dola-seed-2.0-pro	1460.2	25,123

Rank	Model	Rating	Votes
31	gpt-5.4-mini-high	1457.2	13,394
32	qwen3.6-max-preview	1456.9	3,943
33	glm-5	1456.7	19,830
34	gpt-5.1-high	1454.6	40,890
35	claude-sonnet-4-5-20250929-thinking-32k	1453.7	65,810
36	claude-sonnet-4-5-20250929	1452.5	63,729
37	ernie-5.0-0110	1450.9	27,567
38	gemma-4-31b	1450.9	5,807
39	kimi-k2.5-thinking	1449.5	26,016
40	ernie-5.0-preview-1203	1449.4	9,766
41	mimo-v2-pro	1448.8	13,965
42	claude-opus-4-1-20250805-thinking-16k	1448.7	49,874
43	gpt-5.3-chat-latest	1448.5	20,938
44	gemini-2.5-pro	1447.6	113,385
45	qwen3.5-397b-a17b	1447.0	21,269
46	claude-opus-4-1-20250805	1446.7	77,438
47	qwen3.6-plus	1446.3	7,413
48	gpt-4.5-preview-2025-02-27	1444.2	14,547
49	chatgpt-4o-latest-20250326	1442.8	82,545
50	glm-4.7	1442.6	12,143
51	gpt-5.2-high	1439.3	36,581
52	deepseek-v4-flash-thinking	1439.1	3,599
53	gemini-3.1-flash-lite-preview	1439.1	22,256
54	gpt-5.1	1438.9	43,521
55	gpt-5.2	1438.5	33,696
56	gemma-4-26b-a4b	1438.3	5,776
57	longcat-flash-chat-2602-exp	1435.4	11,859
58	qwen3-max-preview	1434.7	27,754
59	gpt-5-high	1433.6	31,979
60	deepseek-v4-flash	1432.1	3,508
61	kimi-k2.5-instant	1431.9	8,196

Subset Rankings (7 models)

Rank	Model	Votes	Overall
1	gemini-3.1-pro-preview	27,941	1492.9
2	gemini-3-pro	41,373	1485.8
3	gemini-3-flash	30,800	1473.6
4	gemini-3-flash (thinking-minimal)	39,847	1462.7
5	gemma-4-31b	5,807	1450.9
6	gemini-3.1-flash-lite-preview	22,256	1439.1
7	gemma-4-26b-a4b	5,776	1438.3

Part 1: General capabilities

Model	overall	coding	math	creative	instruct	hard
gemini-3.1-pro-preview	1492.9	1529.3	1506.9	1489.1	1488.4	1513.7
gemini-3-pro	1485.8	1518.5	1478.2	1485.7	1474.0	1504.0
gemini-3-flash	1473.6	1509.1	1476.3	1459.1	1458.3	1493.2
gemini-3-flash (thinking-minimal)	1462.7	1499.1	1457.0	1450.0	1446.7	1480.5
gemma-4-31b	1450.9	1498.2	1467.7	1422.0	1452.3	1473.5
gemini-3.1-flash-lite-preview	1439.1	1461.4	1436.9	1420.9	1411.7	1448.6
gemma-4-26b-a4b	1438.3	1480.8	1469.2	1404.2	1438.1	1461.2

Part 2: Conversational and industry categories

Model	multi_turn	ind_math
gemini-3.1-pro-preview	1501.9	1496.8
gemini-3-pro	1495.0	1481.3
gemma-4-26b-a4b	1446.4	1476.4
gemini-3-flash	1483.5	1471.7
gemma-4-31b	1461.6	1471.0
gemini-3-flash (thinking-minimal)	1477.7	1458.2
gemini-3.1-flash-lite-preview	1446.6	1432.2

Win/Loss Summary

vs Model	gemini-3.1-pro-preview Wins	Opponent Wins	Overall Gap
gemma-4-26b-a4b	8	0	+54.5
gemini-3.1-flash-lite-preview	8	0	+53.8
gemma-4-31b	8	0	+41.9
gemini-3-flash (thinking-minimal)	8	0	+30.2
gemini-3-flash (thinking-minimal)	8	0	+23.1
gemini-3-flash	8	0	+19.2
gemini-3-pro	8	0	+7.1
gemini-3-flash	8	0	+12.2

Head-to-Head: gemini-3.1-pro-preview vs gemini-3-pro

gemini-3.1-pro-preview wins 8 of 8 categories.

Category	gemini-3.1-pro-preview	gemini-3-pro	Delta	Winner
math	1506.9	1478.2	+28.7	gemini-3.1-pro-preview
ind_math	1496.8	1481.3	+15.4	gemini-3.1-pro-preview
instruct	1488.4	1474.0	+14.4	gemini-3.1-pro-preview
coding	1529.3	1518.5	+10.8	gemini-3.1-pro-preview
hard	1513.7	1504.0	+9.7	gemini-3.1-pro-preview
overall	1492.9	1485.8	+7.1	gemini-3.1-pro-preview
multi_turn	1501.9	1495.0	+6.9	gemini-3.1-pro-preview

Category	gemini-3.1-pro-preview	gemini-3-pro	Delta	Winner
creative	1489.1	1485.7	+3.4	gemini-3.1-pro-preview

Head-to-Head: gemini-3.1-pro-preview vs gemini-3-flash

gemini-3.1-pro-preview wins 8 of 8 categories.

Category	gemini-3.1-pro-preview	gemini-3-flash	Delta	Winner
math	1506.9	1476.3	+30.7	gemini-3.1-pro-preview
instruct	1488.4	1458.3	+30.0	gemini-3.1-pro-preview
creative	1489.1	1459.1	+29.9	gemini-3.1-pro-preview
ind_math	1496.8	1471.7	+25.1	gemini-3.1-pro-preview
hard	1513.7	1493.2	+20.4	gemini-3.1-pro-preview
coding	1529.3	1509.1	+20.2	gemini-3.1-pro-preview
overall	1492.9	1473.6	+19.2	gemini-3.1-pro-preview
multi_turn	1501.9	1483.5	+18.3	gemini-3.1-pro-preview

Head-to-Head: gemini-3.1-pro-preview vs gemini-3-flash (thinking-minimal)

gemini-3.1-pro-preview wins 8 of 8 categories.

Category	gemini-3.1-pro-preview	gemini-3-flash (thinking-minimal)	Delta	Winner
math	1506.9	1457.0	+50.0	gemini-3.1-pro-preview
instruct	1488.4	1446.7	+41.7	gemini-3.1-pro-preview
creative	1489.1	1450.0	+39.0	gemini-3.1-pro-preview
ind_math	1496.8	1458.2	+38.6	gemini-3.1-pro-preview
hard	1513.7	1480.5	+33.2	gemini-3.1-pro-preview
overall	1492.9	1462.7	+30.2	gemini-3.1-pro-preview
coding	1529.3	1499.1	+30.2	gemini-3.1-pro-preview
multi_turn	1501.9	1477.7	+24.2	gemini-3.1-pro-preview

Head-to-Head: gemini-3.1-pro-preview vs gemma-4-31b

gemini-3.1-pro-preview wins 8 of 8 categories.

Category	gemini-3.1-pro-preview	gemma-4-31b	Delta	Winner
creative	1489.1	1422.0	+67.0	gemini-3.1-pro-preview
overall	1492.9	1450.9	+41.9	gemini-3.1-pro-preview
multi_turn	1501.9	1461.6	+40.3	gemini-3.1-pro-preview

Category	gemini-3.1-pro-preview	gemma-4-31b	Delta	Winner
hard	1513.7	1473.5	+40.2	gemini-3.1-pro-preview
math	1506.9	1467.7	+39.2	gemini-3.1-pro-preview
instruct	1488.4	1452.3	+36.1	gemini-3.1-pro-preview
coding	1529.3	1498.2	+31.1	gemini-3.1-pro-preview
ind_math	1496.8	1471.0	+25.8	gemini-3.1-pro-preview

Head-to-Head: gemini-3.1-pro-preview vs gemini-3.1-flash-lite-preview

gemini-3.1-pro-preview wins 8 of 8 categories.

Category	gemini-3.1-pro-preview	gemini-3.1-flash-lite-preview	Delta	Winner
instruct	1488.4	1411.7	+76.7	gemini-3.1-pro-preview
math	1506.9	1436.9	+70.0	gemini-3.1-pro-preview
creative	1489.1	1420.9	+68.2	gemini-3.1-pro-preview
coding	1529.3	1461.4	+68.0	gemini-3.1-pro-preview
hard	1513.7	1448.6	+65.1	gemini-3.1-pro-preview
ind_math	1496.8	1432.2	+64.6	gemini-3.1-pro-preview
multi_turn	1501.9	1446.6	+55.3	gemini-3.1-pro-preview
overall	1492.9	1439.1	+53.8	gemini-3.1-pro-preview

Head-to-Head: gemini-3.1-pro-preview vs gemma-4-26b-a4b

gemini-3.1-pro-preview wins 8 of 8 categories.

Category	gemini-3.1-pro-preview	gemma-4-26b-a4b	Delta	Winner
creative	1489.1	1404.2	+84.9	gemini-3.1-pro-preview
multi_turn	1501.9	1446.4	+55.5	gemini-3.1-pro-preview
overall	1492.9	1438.3	+54.5	gemini-3.1-pro-preview
hard	1513.7	1461.2	+52.5	gemini-3.1-pro-preview
instruct	1488.4	1438.1	+50.3	gemini-3.1-pro-preview
coding	1529.3	1480.8	+48.5	gemini-3.1-pro-preview
math	1506.9	1469.2	+37.7	gemini-3.1-pro-preview
ind_math	1496.8	1476.4	+20.4	gemini-3.1-pro-preview

Head-to-Head: gemini-3-pro vs gemini-3-flash

gemini-3-pro wins 8 of 8 categories.

Category	gemini-3-pro	gemini-3-flash	Delta	Winner
creative	1485.7	1459.1	+26.5	gemini-3-pro
instruct	1474.0	1458.3	+15.7	gemini-3-pro
overall	1485.8	1473.6	+12.2	gemini-3-pro
multi_turn	1495.0	1483.5	+11.4	gemini-3-pro
hard	1504.0	1493.2	+10.7	gemini-3-pro
ind_math	1481.3	1471.7	+9.6	gemini-3-pro
coding	1518.5	1509.1	+9.4	gemini-3-pro
math	1478.2	1476.3	+2.0	gemini-3-pro

Head-to-Head: gemini-3-pro vs gemini-3-flash (thinking-minimal)

gemini-3-pro wins 8 of 8 categories.

Category	gemini-3-pro	gemini-3-flash	Delta	Winner
		(thinking-minimal)		
creative	1485.7	1450.0	+35.6	gemini-3-pro
instruct	1474.0	1446.7	+27.3	gemini-3-pro
hard	1504.0	1480.5	+23.5	gemini-3-pro
overall	1485.8	1462.7	+23.1	gemini-3-pro
ind_math	1481.3	1458.2	+23.1	gemini-3-pro
math	1478.2	1457.0	+21.3	gemini-3-pro
coding	1518.5	1499.1	+19.4	gemini-3-pro
multi_turn	1495.0	1477.7	+17.3	gemini-3-pro

Arena Key Findings

1. **Gemini 3 dominance is comprehensive:** gemini-3.1-pro-preview sweeps all 8 evaluated categories in every one of its 8 head-to-head matchups — a clean 64-0 record across all category comparisons. This isn’t just a top-line rating advantage; it holds across coding, math, creative writing, instruction following, hard prompts, multi-turn, and industry math. Even against the strongest Gemma opponent (gemma-4-31b at rank 38), the narrowest single-category gap is still +10.5 points in math. This level of categorical consistency places gemini-3.1-pro-preview among the most well-rounded models on Arena — it has no exploitable weakness relative to its intra-family competition.
2. **Creative writing is where the gap widens most:** The largest advantages appear in the **creative** category — gemini-3.1-pro-preview leads by +84.9 points over gemma-4-26b-a4b and +67.0 over gemma-4-31b, both representing clear separation by CI analysis. For context, the average creative delta across all matchups is +44.3 points. This suggests Gemini 3’s RLHF tuning or preference optimization invested heavily in subjective quality dimensions. By contrast, **coding** (+31.1 vs gemma-4-31b) and **math** (+25.8 vs gemma-4-26b-a4b at the narrowest) show smaller but still statistically significant gaps — the Gemma models are comparatively more competitive in technical categories.
3. **Gemini 3 has a humanities-leaning quality profile:** Across matchups, the average delta in humanities-oriented categories (creative, instruct) is +40.4 points, compared to +30.8 in STEM categories (coding, math, ind_math). The Gemma models show the inverse pattern — their strongest relative showings are consistently in math and coding. This divergence may reflect different training emphasis: Gemini 3 appears optimized for the kind of open-ended, stylistically demanding tasks that drive Arena votes, while Gemma 4 prioritizes technical correctness per parameter.

4. **Tier stratification within the lineup is clear:** The 7 evaluated models span from Frontier (rank 4) to Upper-mid (rank 56), a range of 54.6 Arena rating points. The Gemini 3 models form a tight cluster in the Frontier/Near-frontier range (ranks 4–26, ratings 1462.7–1492.9), while the Gemma 4 models sit in the Upper-mid tier (ranks 38 and 56, ratings 1438.3–1450.9). The gap between the lowest Gemini model (gemini-3-flash thinking-minimal at 1462.7) and the highest Gemma model (gemma-4-31b at 1450.9) is 11.8 points — a small but statistically significant difference that confirms these are genuinely distinct performance tiers rather than noise.
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Part 2: Artificial Analysis

Source: Artificial Analysis

About Artificial Analysis

Artificial Analysis evaluates models using **automated benchmark suites** – standardized tests with known correct answers, run programmatically. Their Intelligence Index (v4.0) aggregates scores from 10 evaluations:

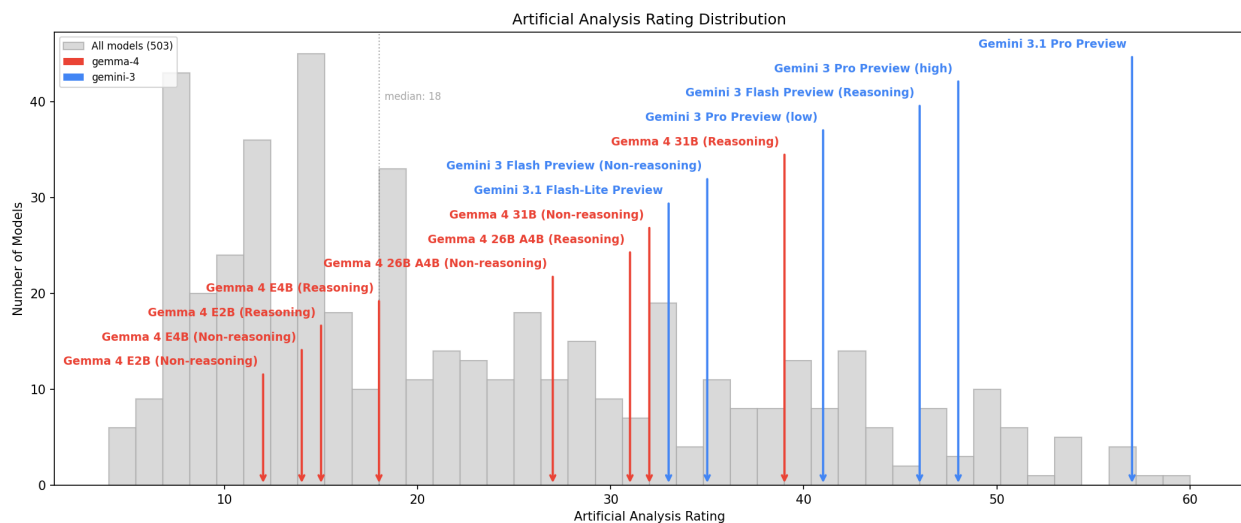
- **GPQA Diamond** – graduate-level science questions
- **Humanity’s Last Exam (HLE)** – extremely difficult cross-domain questions
- **SciCode** – scientific coding problems
- **Terminal-Bench Hard** – complex terminal/CLI tasks
- **IFBench** – instruction following
- **AA-LCR** – long-context retrieval
- **AA-Omniscience** – broad knowledge assessment
- **GDPval-AA** – GDP prediction (quantitative reasoning)
- **tau2-Bench Telecom** – domain-specific agent tasks
- **CritPt** – critical thinking

Unlike Arena’s human preference votes, these benchmarks have **objectively correct answers**. This makes AA scores more precise for measurable capabilities (coding, math, factual recall) but less reflective of subjective qualities like writing style, helpfulness, or conversational fluency.

AA also independently measures **speed** (output tokens/sec), **latency** (time to first token), and **pricing** across API providers, providing a practical deployment perspective.

Models evaluated: Gemini 3 Flash Preview (Non-reasoning), Gemini 3.1 Pro Preview, Gemini 3 Pro Preview (low), Gemini 3.1 Flash-Lite Preview, Gemini 3 Flash Preview (Reasoning), Gemini 3 Pro Preview (high), Gemma 4 26B A4B (Reasoning), Gemma 4 31B (Reasoning), Gemma 4 26B A4B (Non-reasoning), Gemma 4 31B (Non-reasoning), Gemma 4 E2B (Non-reasoning), Gemma 4 E4B (Reasoning), Gemma 4 E4B (Non-reasoning), Gemma 4 E2B (Reasoning)

Global AA Rankings (503 models total)



Rank	Model	AA Intelligence
1	GPT-5.5 (xhigh)	60
2	GPT-5.5 (high)	58
3	Gemini 3.1 Pro Preview	57
4	Claude Opus 4.7 (Adaptive Reasoning, Max Effort)	57
5	GPT-5.5 (medium)	56
6	GPT-5.4 (xhigh)	56
7	GPT-5.3 Codex (xhigh)	53
8	Grok 4.3	53
<i>[16 models not shown]</i>		
25	GLM-5 (Reasoning)	49
26	GPT-5.2 Codex (xhigh)	49
27	Claude Opus 4.5 (Reasoning)	49
28	Grok 4.20 0309 v2 (Reasoning)	49
29	GPT-5.4 mini (xhigh)	48
30	Gemini 3 Pro Preview (high)	48
31	Grok 4.20 0309 (Reasoning)	48
32	GPT-5.4 (low)	47
33	GPT-5.1 (high)	47
34	Gemini 3 Flash Preview (Reasoning)	46
35	DeepSeek V4 Flash (Reasoning, Max Effort)	46
36	GLM-5-Turbo	46
37	GPT-5.2 (medium)	46
38	Claude Opus 4.6 (Non-reasoning, High Effort)	46
39	Kimi K2.5 (Reasoning)	46
<i>[17 models not shown]</i>		
57	GLM 5V Turbo (Reasoning)	42
58	GPT-5 (medium)	42
59	Claude 4.1 Opus (Reasoning)	42
60	GLM-4.7 (Reasoning)	42
61	Qwen3.5 27B (Reasoning)	42
62	Gemini 3 Pro Preview (low)	41
63	MiMo-V2-Flash (Feb 2026)	41

Rank	Model	AA Intelligence
64	Hy3-preview (Reasoning)	41
65	Qwen3.5 122B A10B (Reasoning)	41
66	GPT-5 mini (high)	41
67	DeepSeek V3.2 (Reasoning)	41
68	Grok 4	41
69	MiniMax-M2.5	41
70	GPT-5.5 (Non-reasoning)	40
71	GLM-5 (Non-reasoning)	40
72	Qwen3.5 397B A17B (Non-reasoning)	40
73	o3-pro	40
74	Kimi K2 Thinking	40
75	Gemma 4 31B (Reasoning)	39
76	Mistral Medium 3.5	39
77	DeepSeek V4 Pro (Non-reasoning)	39
78	GPT-5 (low)	39
79	Claude 4 Opus (Reasoning)	39
80	MiniMax-M2.1	39
	<i>[19 models not shown]</i>	
100	NVIDIA Nemotron 3 Super 120B A12B (Reasoning)	36
101	KAT-Coder-Pro V1	36
102	Claude 4.1 Opus (Non-reasoning)	36
103	MiniMax-M2	36
104	GPT-5.4 (Non-reasoning)	35
105	Gemini 3 Flash Preview (Non-reasoning)	35
106	Nova 2.0 Pro Preview (medium)	35
107	MiMo-V2.5-Pro (Non-reasoning)	35
108	Qwen3.5 122B A10B (Non-reasoning)	35
109	Grok 4 Fast (Reasoning)	35
110	Gemini 2.5 Pro	34
111	Nova 2.0 Lite (high)	34
112	Claude 3.7 Sonnet (Reasoning)	34
113	GLM-4.7 (Non-reasoning)	34
114	gpt-oss-120B (high)	33
115	Gemini 3.1 Flash-Lite Preview	33
116	Hy3-preview (Non-reasoning)	33
117	Ling-2.6-1T	33
118	Doubao Seed Code	33
119	o4-mini (high)	33
120	GPT-5.2 (Non-reasoning)	33
121	Claude 4 Sonnet (Non-reasoning)	33
122	Claude 4 Opus (Non-reasoning)	33
123	DeepSeek V3.1 Terminus (Reasoning)	33
124	Gemma 4 31B (Non-reasoning)	32
125	Grok 3 mini Reasoning (high)	32
126	Mercury 2	32
127	K-EXAONE (Reasoning)	32
128	Qwen3.5 9B (Reasoning)	32
129	DeepSeek V3.2 (Non-reasoning)	32
130	DeepSeek V3.2 Exp (Reasoning)	32
131	GLM-4.6 (Reasoning)	32
132	Qwen3 Max Thinking (Preview)	32
133	Gemma 4 26B A4B (Reasoning)	31

Rank	Model	AA Intelligence
134	Claude 4.5 Haiku (Non-reasoning)	31
135	Nova 2.0 Pro Preview (low)	31
136	Trinity Large Thinking	31
137	Qwen3.6 35B A3B (Non-reasoning)	31
138	Gemini 2.5 Flash Preview (Sep '25) (Reasoning)	31
	<i>[20 models not shown]</i>	
159	Qwen3 Coder Next	28
160	DeepSeek V3.1 (Non-reasoning)	28
161	DeepSeek V3.2 Exp (Non-reasoning)	28
162	DeepSeek V3.1 Terminus (Non-reasoning)	28
163	Apriel-v1.5-15B-Thinker	28
164	Gemma 4 26B A4B (Non-reasoning)	27
165	Magistral Medium 1.2	27
166	Mistral Small 4 (Reasoning)	27
167	DeepSeek R1 0528 (May '25)	27
168	Apriel-v1.6-15B-Thinker	27
169	Qwen3.5 4B (Reasoning)	27
	<i>[77 models not shown]</i>	
247	Gemini 2.0 Flash Thinking Experimental (Jan '25)	19
248	Qwen3 235B A22B (Reasoning)	19
249	Qwen3 VL 30B A3B (Reasoning)	19
250	QwQ 32B	19
251	Llama 4 Maverick	18
252	Gemma 4 E4B (Reasoning)	18
253	Mistral Small 4 (Non-reasoning)	18
254	Magistral Small 1.2	18
255	Nova 2.0 Lite (Non-reasoning)	18
256	Llama Nemotron Super 49B v1.5 (Reasoning)	18
257	Llama 3.3 Nemotron Super 49B v1 (Reasoning)	18
	<i>[40 models not shown]</i>	
298	Qwen2.5 Max	16
299	Qwen3 VL 30B A3B Instruct	16
300	Qwen3 32B (Reasoning)	16
301	Qwen3 VL 8B (Reasoning)	16
302	Qwen3 14B (Reasoning)	16
303	Gemma 4 E2B (Reasoning)	15
304	Falcon-H1R-7B	15
305	Llama 3.1 Nemotron Ultra 253B v1 (Reasoning)	15
306	Step3 VL 10B	15
307	ERNIE 4.5 300B A47B	15
308	Qwen3 Omni 30B A3B (Reasoning)	15
309	Ling-flash-2.0	15
310	Claude 3.5 Sonnet (Oct '24)	15
311	Mistral Large 2 (Nov '24)	15
312	Devstral Small (Jul '25)	15
313	Mistral Small 3.2	15
314	DeepSeek R1 Distill Qwen 14B	15
315	Sonar	15
316	Sonar Pro	15
317	GLM-4.5V (Reasoning)	15
318	Qwen2.5 Instruct 72B	15
319	Qwen3 30B A3B 2507 Instruct	15

Rank	Model	AA Intelligence
320	Qwen3 30B A3B (Reasoning)	15
321	QwQ 32B-Preview	15
322	Llama 3.3 Instruct 70B	14
323	Gemma 4 E4B (Non-reasoning)	14
324	Minstral 3 8B	14
325	Solar Pro 2 (Reasoning)	14
326	Llama Nemotron Super 49B v1.5 (Non-reasoning)	14
327	NVIDIA Nemotron Nano 9B V2 (Reasoning)	14
328	NVIDIA Nemotron Nano 12B v2 VL (Reasoning)	14
	<i>[32 models not shown]</i>	
361	Grok 2 (Dec '24)	13
362	Nova Pro	13
363	Qwen3 8B (Reasoning)	13
364	Qwen3 VL 4B (Reasoning)	13
365	Qwen2.5 Instruct 32B	13
366	Gemma 4 E2B (Non-reasoning)	12
367	R1 1776	12
368	Olmo 3.1 32B Instruct	12
369	Granite 4.1 8B	12
370	Hermes 4 - Llama-3.1 70B (Non-reasoning)	12
371	Sarvam 30B (high)	12

Reasoning Models

Model	Params (total/active)	AA Intelligence	Coding	Math	Speed (t/s)	TTFT (s)	Price (\$/1M blend)	Context
Gemini 3 Pro Preview (high)	proprietary	48	46	95	126.5	19.97	\$4.50	–
Gemini 3 Flash Preview (Reasoning)	proprietary	46	42	97	179.9	5.46	\$1.12	–
Gemini 3 Pro Preview (low)	proprietary	41	39	86	–	–	\$4.50	–
Gemma 4 31B (Reasoning)	proprietary	39	38	–	35.3	1.00	\$0.20	–
Gemini 3 Flash Preview (Non-reasoning)	proprietary	35	37	55	185.2	0.70	\$1.12	–
Gemma 4 31B (Non-reasoning)	proprietary	32	33	–	–	–	–	–

Model	Params (total/active)	AA Intelligence	Coding	Math	Speed (t/s)	TTFT (s)	Price (\$/1M blend)	Context
Gemma 4 26B A4B (Reasoning)	proprietary	31	22	–	–	–	\$0.20	–
Gemma 4 26B A4B (Non-reasoning)	proprietary	27	29	–	–	–	–	–
Gemma 4 E4B (Reasoning)	proprietary	18	13	–	–	–	–	–
Gemma 4 E2B (Reasoning)	proprietary	15	9	–	–	–	–	–
Gemma 4 E4B (Non-reasoning)	proprietary	14	6	–	–	–	–	–
Gemma 4 E2B (Non-reasoning)	proprietary	12	8	–	–	–	–	–

Non-Reasoning Models

Model	Params (total/active)	AA Intelligence	Coding	Speed (t/s)	TTFT (s)	Price (\$/1M blend)	Context
Gemini 3.1 Pro Preview	proprietary	57	55	129.9	21.80	\$4.50	–
Gemini 3.1 Flash-Lite Preview	proprietary	33	30	310.4	4.93	\$0.56	–

All Models

Model	AA Intelligence	Coding	Math	Speed (t/s)	TTFT (s)	Price (\$/1M blend)	Context
Gemini 3.1 Pro Preview	57	55	–	129.9	21.80	\$4.50	–
Gemini 3 Pro Preview (high)	48	46	95	126.5	19.97	\$4.50	–
Gemini 3 Flash Preview (Reasoning)	46	42	97	179.9	5.46	\$1.12	–
Gemini 3 Pro Preview (low)	41	39	86	–	–	\$4.50	–
Gemma 4 31B (Reasoning)	39	38	–	35.3	1.00	\$0.20	–

Model	AA Intelligence	Coding	Math	Speed (t/s)	TTFT (s)	Price (\$/1M blend)	Context
Gemini 3 Flash Preview (Non-reasoning)	35	37	55	185.2	0.70	\$1.12	—
Gemini 3.1 Flash-Lite Preview	33	30	—	310.4	4.93	\$0.56	—
Gemma 4 31B (Non-reasoning)	32	33	—	—	—	—	—
Gemma 4 26B A4B (Reasoning)	31	22	—	—	—	\$0.20	—
Gemma 4 26B A4B (Non-reasoning)	27	29	—	—	—	—	—
Gemma 4 E4B (Reasoning)	18	13	—	—	—	—	—
Gemma 4 E2B (Reasoning)	15	9	—	—	—	—	—
Gemma 4 E4B (Non-reasoning)	14	6	—	—	—	—	—
Gemma 4 E2B (Non-reasoning)	12	8	—	—	—	—	—

Artificial Analysis Key Findings

- Gemini 3.1 Pro Preview is a Frontier-tier model on automated benchmarks:** With an Intelligence Index of 57 and rank 3 of 503 models globally, it sits behind only GPT-5.5 (xhigh and high configurations) and alongside Claude Opus 4.7 at rank 4. This confirms the Arena picture — Gemini 3’s flagship competes at the very top across both human-preference and automated evaluation methodologies. Its coding sub-score (55) is similarly elite, and its math score of 95 (on the Pro high-reasoning variant) is among the highest on the platform.
- The Gemini-to-Gemma quality gap is dramatic on automated benchmarks:** Gemini 3.1 Pro Preview scores 57 while the best Gemma model (Gemma 4 31B Reasoning) scores 39 — a clear separation of 18 Intelligence Index points, well over 1.0 population standard deviations. The gap widens further down the Gemma lineup: the smaller E2B and E4B models score 12–18, placing them in the Long-tail tier. This spread illustrates that Gemma 4 is less a single competitive model and more a family spanning from competitive (31B) to entry-level (E2B).
- Gemma 4 offers an exceptional cost-to-quality ratio:** At \$0.20/1M blended tokens, the Gemma 4 31B (Reasoning) delivers Intelligence Index 39 — Upper-mid tier quality at roughly 1/22nd the price of Gemini 3 Pro (\$4.50) and 1/5.6th the price of Gemini 3 Flash (\$1.12). For applications where Upper-mid quality suffices and cost is the binding constraint, this is a compelling tradeoff. The 31B model’s low latency (1.00s TTFT) also makes it viable for interactive applications despite its slower throughput (35.3 t/s vs Gemini Flash’s 179.9 t/s).
- Speed and infrastructure maturity favor Gemini 3:** Gemini Flash variants deliver 179.9–310.4

t/s, while even the fastest Gemma model (31B Reasoning) manages only 35.3 t/s — roughly 5x slower. Many Gemma configurations lack speed and latency data entirely on AA, suggesting they may not yet be widely deployed through optimized inference providers. This is consistent with Gemma’s positioning as a self-hosted open-weights model rather than a managed API product.

Definitions

Model Tiers

Tiers are based on absolute rank within each data source:

Tier	Rank Range
Frontier	1–10
Near-frontier	11–25
Upper-mid	26–75
Mid-tier	76–150
Long-tail	151+

Gap Significance

Arena — uses per-model confidence intervals (CI). If CIs overlap, the models are “statistically indistinguishable.” If CIs don’t overlap but the closest bounds are within 20 points, the difference is “small but statistically significant.” A gap of 20+ points between closest bounds indicates “clear separation.”

Artificial Analysis — uses gap relative to population standard deviation. A gap below 0.5 stdev is “not clearly distinguishable,” 0.5–1.0 stdev indicates a “moderate difference,” and above 1.0 stdev indicates “clear separation.”

Data provided by *Artificial Analysis* (<https://artificialanalysis.ai>) and *Arena* (<https://lmarena.ai>).